Meghan J. Blumstein

509 Lowell Mail Center • Cambridge, MA 02138 • (610) 331-6584
mblumste@mit.edu • blumsteinm.github.io
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I am an interdisciplinary plant biologist using tools from ecophysiology, genetics & genomics, remote sensing and predictive modeling to answer the question "how will temperate trees survive predicted global change?"

EDUCATION

Harvard University | Ph.D.
Organismic and Evolutionary Biology | Cambridge, MA — August 2020
Dissertation Committee: Missy Holbrook (advisor), Robin Hopkins, Elena Kramer, Johnathan Thompson
In Collaboration with: The Plant Systems Biology Group, Oak Ridge National Laboratory, DOE

Middlebury College | B.A. with Honors
Joint Degree in Biology and Environmental Studies | Middlebury, VT — May 2011
Thesis Committee: Andrea Lloyd, Stephen Trombulak, Matt Landis

RESEARCH POSITIONS

NSF Postdoctoral Researcher | MIT / CEFE (France) — 2020-Present
Combine environmental, genomic and phenomic information from Red oak (Quercus rubra) into a unified, process-based model of phenological timing.

Ph.D. Candidate | Harvard University — 2014 – 2020
Detailed the genetic and environmental drivers as well as the physiological costs of sugar storage in black cottonwood (Populus trichocarpa). Used genomic data in a novel way to make novel predictions of species survival. Funded by NSF GRFP and DOE SCGSR fellowships.

Research Assistant Thompson Lab| Harvard Forest — 2014
Research Assistant Thompson Lab| Smithsonian Conservation Biology Institute — 2013
Programmed large ecosystem service models using python and ArcGIS and analyzed outputs in R for large NSF-funded project. Performed independent project on ecosystem service hotspots and published results.

Fellow | National Trust (Cornwall, UK) — 2012
Worked with the Head of Plant Pest and Disease for the Trust, Ian Wright, to assess gardener's understanding of and preparedness for dealing with invasive pests and disease.

Research Assistant Lloyd Lab | Middlebury College — 2011
Worked in a tree-ring lab measuring and cross-dating black spruce (Picea mariana) for project evaluating the likelihood of warming-induced regime shifts from conifer forest to steppe.

TEACHING EXPERIENCE

Fellowship Writing and Peer Review Workshop — 2018 & 2019
Harvard Forest REU program students

R Programming Series — 2019 & 2020

Harvard OEB REU program students

The Green Planet: Plants that Changed Human History 2019
Harvard Extension School: ANTH E-1145

Ecology and the Design World 2018
Harvard Graduate School of Design: SCI 6241

Genetics and Genomics 2018
Harvard College: OEB 50

Plant Biology 2017
Harvard College: OEB 52

7th Grade Environmental Science 2010
9th Grade Introductory Biology
New Canaan Country School

PUBLICATIONS

Blumstein, M., R. Hopkins. *In Submission PC&E*. Adaptive variation and plasticity in nonstructural carbohydrate storage in a temperate tree species. Preprint on Authorea DOI: 10.22541/au.160071423.34337856

Blumstein, M., A. Sala, D. Weston, N.M. Holbrook, R. Hopkins. *In Submission*. Trees actively allocate carbohydrates to storage.

Blumstein, M., A.D. Richardson, D. Weston, J. Zhang, W. Muchero, and R. Hopkins. 2020. *STAR Protocols*. Protocol for projecting allele frequency change under future climate change at adaptive-associated loci.

Blumstein, M., A.D. Richardson, D. Weston, J. Zhang, W. Muchero, and R. Hopkins. 2020. *Current Biology*. A new perspective on ecological prediction reveals limits to climate adaptation in a temperate tree species.

Blumstein, M. and J.R. Thompson. 2015 *Journal of Applied Ecology*. Land use impacts on the quantity and configuration of ecosystem service provisioning in Massachusetts, USA. .

Thompson J. R., K. Fallon-Lambert, D.R. Foster, E. Broadbent**, M. Blumstein**, A. Almeyda Zambrano, Y. Fan. 2016. *Ecosphere*. The consequences of four plausible land-use scenarios for forest ecosystems and their services in Massachusetts, USA. .

White Paper: Changes to the Land: Four Scenarios for the Future of the Massachusetts Landscape. Harvard Forest, Harvard University. ISBN:9780615985268

SELECT POSTERS & PRESENTATIONS

Blumstein, M. Can variation in sugar storage enable temperate trees to survive climate change? 2020. *Environmental Science Seminar: MIT*. **Talk**

Blumstein, M., D. Weston, R. Hopkins, N.M. Holbrook. Resolving a hypothesized tradeoff between the active allocation of carbon to NSC storage versus other sink demands. 2020. *ESA Annual Meeting.* **Talk**

Barker-Plotkin, A.A.B., **M. Blumstein**, D. Laflower, S. Kruszka, J. Elkinton, V. Pasquarella, B. Bradley, J.R. Thompson. Do defoliated oaks die from carbon starvation? 2020. *ESA Annual Meeting*. **Talk**

Blumstein, M, A.D. Richardson, D. Weston, J. Zhang, W. Muchero, and R. Hopkins. Current Biology. A new perspective on ecological prediction reveals limits to climate adaptation in a temperate tree species. 2020. *Society of Comparative Biology (SICB),* 2020. #15-2.**Talk**

Blumstein, M., D. Weston, A. Richardson, R. Holbrook. A novel approach to quantifying the drivers of alternate carbon allocation strategies in a temperate, deciduous tree species. 2018*. American Geophysical Union (AGU)*, Fall Meeting 2018, abstract #GC338-08. **Talk**

Blumstein, M., R Hopkins, D J Weston, N M Holbrook, A D Richardson. Local adaptation of root sugar storage in black cottonwood (*P. trichocarpa*). *Multiscale Plant Vascular Biology GRC*. 2018. Mt. Snow, Vermont. **Poster**

Lloyd, A., P. Duffy, D. Mann, M. Leonawicz, **M. Blumstein**, and E. Pendall. 2011. Threshold responses of aspen and spruce growth to temperature may presage a regime shift in the boreal forest. American Geophysical Union, Fall Meeting 2011, abstract #U51-05. **Talk**

GRANTS AND AWARDS

NSF Postdoctoral Research Fellowship ($216,000) 2020
National Science Foundation
Covers salary and research for 3 years

RAPID: Is Carbon Starvation a Proximal Cause of Tree Mortality from Defoliation? ($99,340) 2019
PI J. R. Thompson (Blumstein is Senior Personnel)
National Science Foundation
Study of recent outbreak of gypsy moths in MA, with aim to build predictive model of tree mortality

Certificate of Distinction in Teaching (OEB 50) 2018
Derek Bok Center (Harvard)
Recognizes teaching fellows with 4.5/5 or higher on course evaluations

Deland Grant ($3,500) 2018
Arnold Arboretum
Grant to support sequencing work for project on Quercus rubrum phenology

Certificate of Distinction in Teaching (OEB 52) 2017
Derek Bok Center (Harvard)
Recognizes teaching fellows with 4.5/5 or higher on course evaluations

SCGSR Fellowship ($15,000) 2017
Department of Energy
Supports fellows working in National Labs

Explorers Club ($2,500) 2016
Student grant in aid of field work

Southwood Prize: runner-up 2015
The Journal of Applied Ecology
Awarded annually to the best paper by an early career researcher

Pierce Fellowship ($18,000) 2014

Harvard University
Awarded annually to "a select group of the most talented prospective PhD students in the natural sciences and engineering"

NSF Graduate Student Fellowship ($132,000) 2014
National Science Foundation
Covers graduate student tuition and stipend for 3 years

The Damaris Horan Prize Fellowship ($10,000) 2012
Royal Oak Foundation/National Trust of England
Awarded to an American each year to bring their unique talents to the National Trust

EDUCATIONAL OUTREACH

NSF REU Program | Cambridge, MA 2018-Present
Program Advisory Committee member: assisted with the writing of OEB's first NSF REU proposal and helped with the logistics of the program prior to arrival and during students' tenure on campus.

Harvard College | Cambridge, MA 2015-Present
Resident Tutor in Lowell House: one of the 12 undergraduate houses containing ~450 students. I serve as a residential advisor, supporting students with their academic, extracurricular, and personal endeavors. Specifically have served as a fellowships advisor, a CARE tutor (Consent Advocate & Relationship Educator), the JCR (Junior Common Room) tutor who helps students plant inclusive community events, and the SCR tutor (Senior Common Room) who organizes hiring and fosters connections between SCR members such as professors and local professionals with students interested in similar paths.

Harvard Forest | Petersham, MA 2014-Present
Science Education Guide: I lead regular tours at Harvard Forest for groups of all ages and interests.

SCIENTIFIC POLICY EXPERIENCE

Scientific Citizenship Initiative (SCI) Fellow 2019
Researched and proposed environmental legislation in Massachusetts State Senator Julian Cyr's office

Analysis of four future scenarios of land use change in Massachusetts 2014
Report to Stakeholders (Land Managers and MA Legislators)

Protecting garden plants from invasive pest and disease 2013
Report to National Trust (UK) and DEFRA

Forest regeneration in Valley Forge National Historical Park 2011
Analysis and recommendation of potential forest management strategies

Analysis of Arsenic Contamination in Private Wells 2010
Report to legislators on arsenic contamination and proposed legislation

MENTORING

Theresa Caso-McHugh | 2020 - Present

Andrew Castillo | 2020 – Present | *Thesis: What is the role of STAMN2 in Alzheimer's Disease?*

Sophie Webster | 2020 – Present | *Thesis: Are we underpredicting the urban heat island effect?*

Aaron Viser | 2016 – 2018

Amy Zhou | 2017

Emma Borjigin-Wang | 2016 – 2017 | *Thesis: How does sugar storage relate to wood anatomy?*

COMPUTATIONAL SKILLS

Fluent (10+ years): R, Python, ArcGIS, LaTeX, Adobe Suite, InVEST
Highly proficient (2-5 years): HTML, CSS, fortran, C/C++, ENVI, Imagine, Landis-II
Certificates: Tower Climbing (2014), Commercial Drone Pilot's License (2020)

REFERENCES

Missy Holbrook
Professor of Organismic and Evolutionary Biology
Charles Bullard Professor of Forestry
Harvard University
Cambridge, MA 02138
noelmichele@gmail.com
617.496.0603

Robin Hopkins
Assistant Professor of Organismic and Evolutionary Biology
Department of Organismic & Evolutionary Biology
Harvard University
Cambridge, MA 02138
rhopkins@fas.harvard.edu
617.384.5501

Johnathan Thompson
Senior Ecologist
Harvard Forest
Petersham, MA 01366
jthomps@fas.harvard.edu
978.724.3302

David Weston
Staff Scientist
Biosciences Division
Oak Ridge National Laboratory (ORNL)
1 Beth Valley Road Bldg. 1507
Oak Ridge, TN 37830
westondj@ornl.gov

865.241.8323